

Math 142, S12

Quiz 1

Section:

Name: Answer

SSN: xxx-xx-

Instructions: There are totally 2 problems. You must show all of your work to receive a credit for a correct answer. No calculator is allowed in the quiz and exam.

Problem 1. (5 points) Evaluate the following indefinite integrals:

(a) $\int \frac{x}{x^2+2} dx$

$$u = x^2 + 2 \Rightarrow du = 2x dx$$

$$\frac{1}{2} du = x dx$$

$$\int \frac{x}{x^2+2} dx \stackrel{u=x^2+2}{=} \int \frac{1}{u} \cdot \frac{1}{2} du$$

$$= \frac{1}{2} \ln|u| + C$$

$$= \frac{1}{2} \ln(x^2+2) + C$$

(b) $\int \cos \theta \sin^3 \theta d\theta$

$$u = \sin \theta \Rightarrow du = \cos \theta d\theta$$

$$\int \cos \theta \sin^3 \theta d\theta \stackrel{u=\sin \theta}{=} \int u^3 du$$

$$= \frac{u^4}{4} + C$$

$$= \frac{\sin^4 \theta}{4} + C$$

Problem 2. (5 points) Evaluate the following definite integrals:

(a) $\int_1^e \frac{(\ln x)^2}{x} dx$

$$u = \ln x, du = \frac{1}{x} dx$$

when $x=1$, $u = \ln 1 = 0$

$x=e$, $u = \ln e = 1$.

Thus

$$\int_1^e \frac{(\ln x)^2}{x} dx \stackrel{u=\ln x}{=} \int_0^1 u^2 du$$

$$= \frac{u^3}{3} \Big|_0^1$$

$$= \frac{1}{3}$$

(b) $\int_0^1 x e^{-x^2} dx$

$$u = -x^2, du = -2x dx$$

$$-\frac{1}{2} du = x dx$$

when $x=0$, $u = -0^2 = 0$

$x=1$, $u = -(1)^2 = -1$

Thus

$$\int_0^1 x e^{-x^2} dx \stackrel{u=-x^2}{=} \int_0^{-1} e^u \left(-\frac{1}{2}\right) du$$

$$= -\frac{1}{2} e^u \Big|_0^{-1}$$

$$= -\frac{1}{2} (e^{-1} - 1)$$

$$= \frac{1}{2} \left(1 - \frac{1}{e}\right)$$